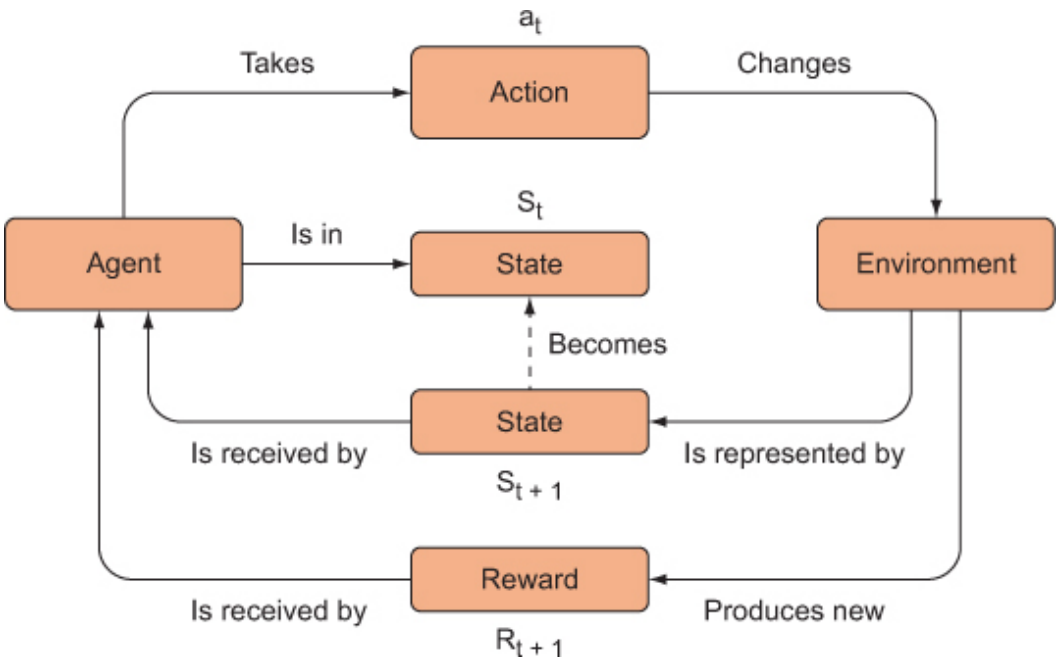
Why would we even think about abandoning backpropagation? Well, with both DQN and policy gradient approaches we created one agent whose policy depended on a neural network to approximate the Q function or policy function. As shown in figure 6.1, the agent interacts with the environment, collects experiences, and then uses backpropagation to improve the accuracy of its neural network and, hence, its policy. We needed to carefully tune several hyperparameters ranging from selecting the right optimizer function, mini-batch size, and learning rate so that the training would be stable and successful. Since the training of both DQN and policy gradient algorithms relies on stochastic gradient descent, which as the name suggests relies on noisy gradients, there is no guarantee that these models will successfully learn (i.e., converge on a good local or global optimum).

Figure 6.1. For the past algorithms that we covered, our agent interacted with environment, collected experiences, and then learned from those experiences. We repeated the same process over and over for each epoch until the agent stopped learning.



Depending on the environment and the complexity of the network, creating an agent with the right hyperparameters may be incredibly difficult. Moreover, in order to use gradient descent and backpropagation, we need a model that is differentiable. There are certainly interesting and useful models you could construct that might be impossible to train with gradient descent due to the lack of differentiability.